

MCL

iRODS Use Case: ALPmat, a Platform for Collaborative AI-driven Advanced Materials Design

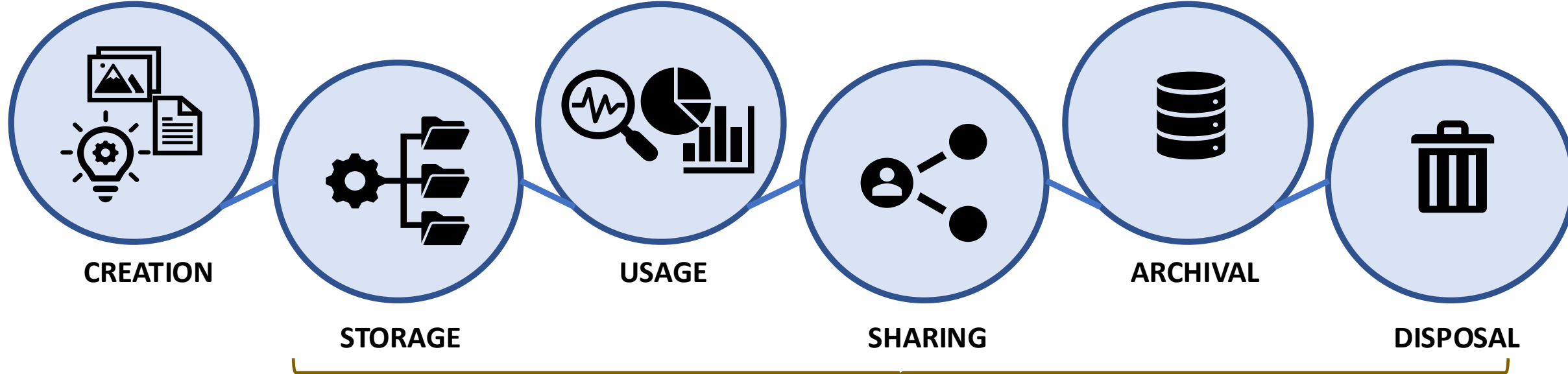


Natalia Bedoya, Bernd Schuscha, Dominik Brandl, Christian Stecher,
Günther Moser, Daniel Scheiber, Jürgen Spitaler
(Materials Center Leoben Forschung GmbH, Leoben Austria)

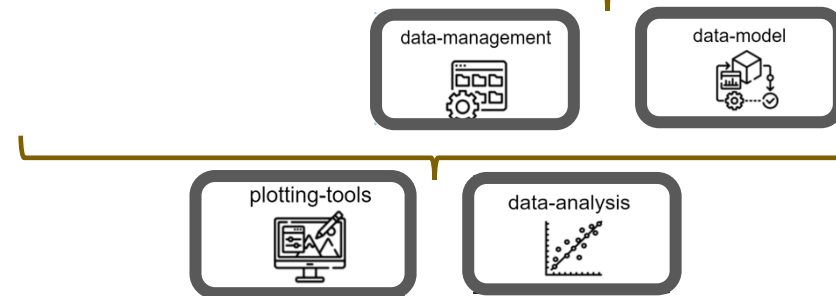
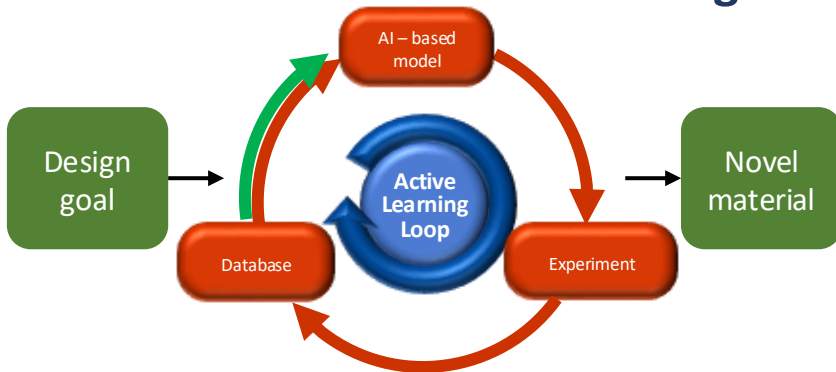
Han Tran, Heimo Gursch (Know Center Research GmbH, Graz, Austria)

Lorenz Romaner (Montanuniversität Leoben, Leoben, Austria)

Data Lifecycle



Framework Accelerated Materials Design



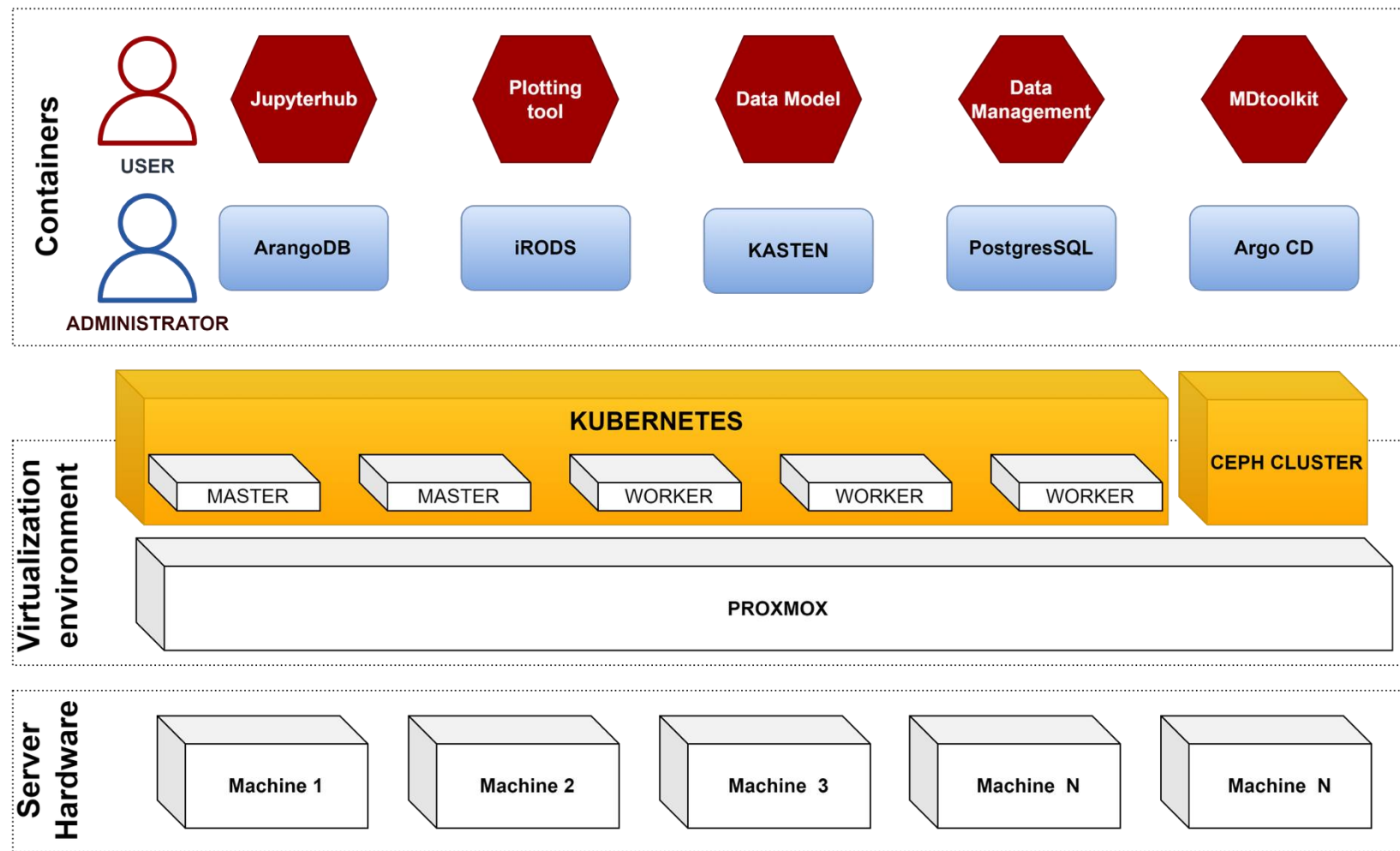
LPmat



LPmat

Key features

- ☐ **Modular and extendible** architecture.
- ☐ Fulfillment of **high security** requirements.
- ☐ **Data sovereignty** guaranteed.
- ☐ **All elements hosted at MCL.**
- ☐ **Easy access via Web-browser**



Data Management

Help

At iRODS and Metalnx UI, you can search for, upload, download, and manage data and metadata. Admin accounts also provide access to documents uploaded from Literature Search Tools and Data Model.

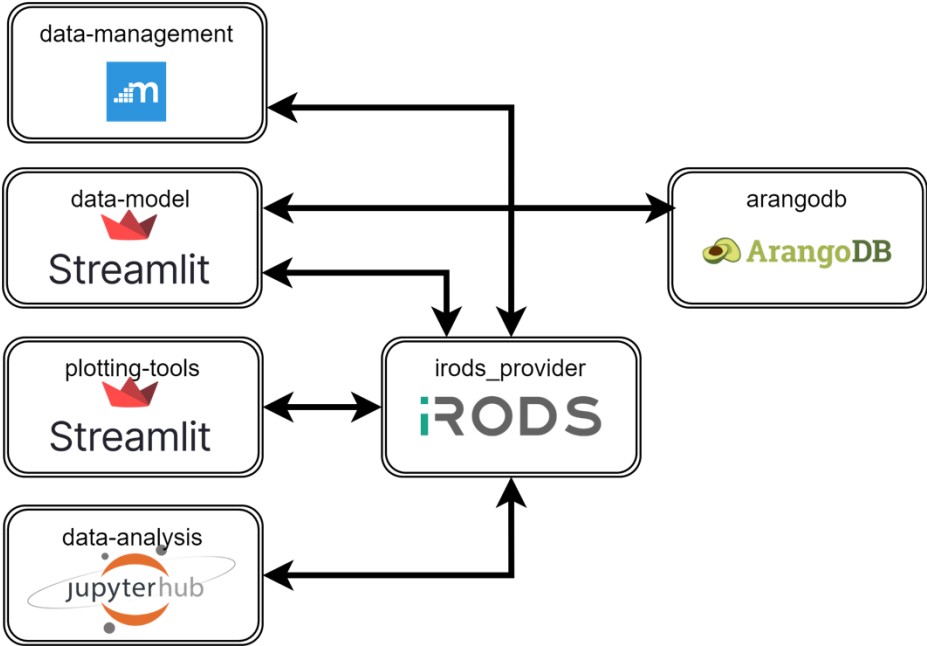
Data management for data and metadata. Users decide storage structure, metadata and permissions.

Data Model

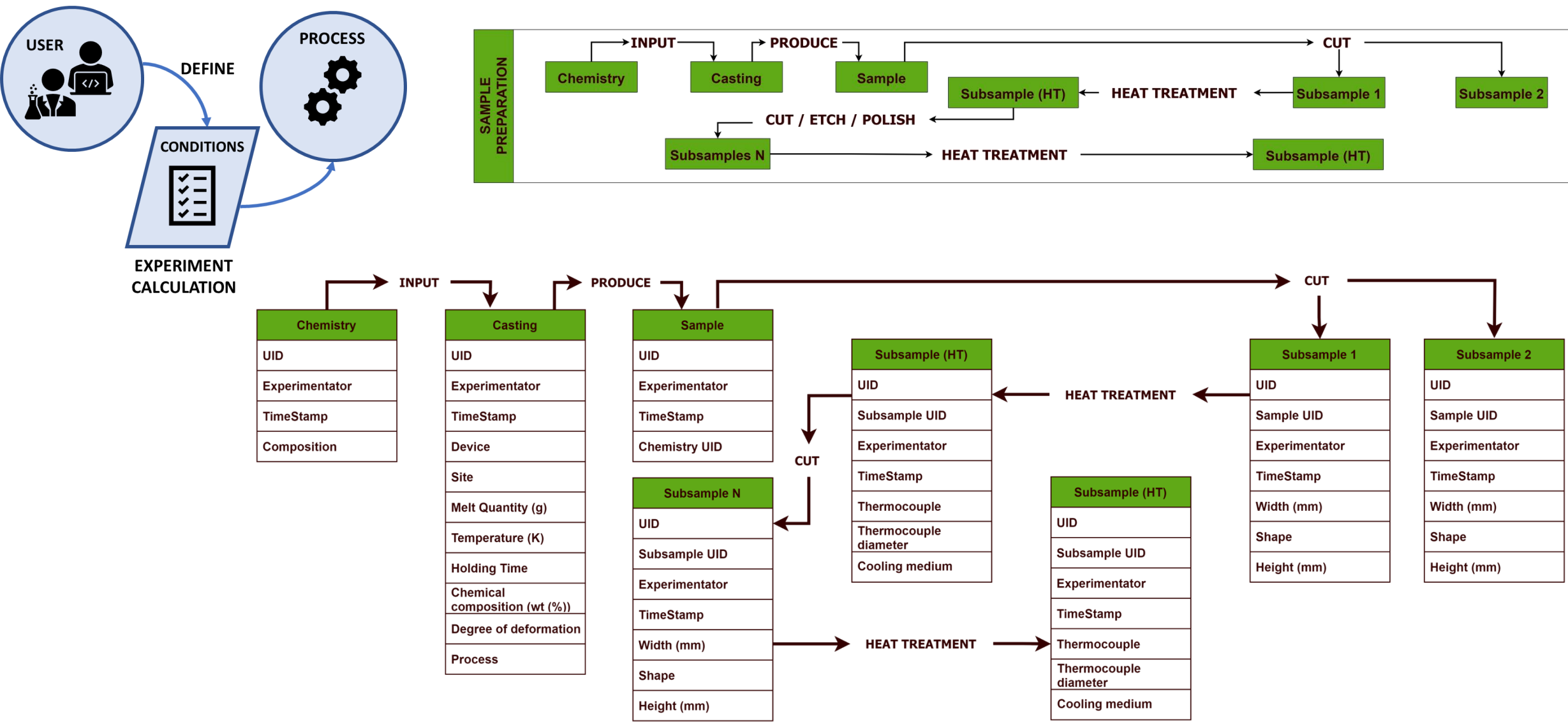
Help

You can input and view the complete workflow of Bainite experiments here.

Platform provides tailored data models to save data and metadata of scientific workflows

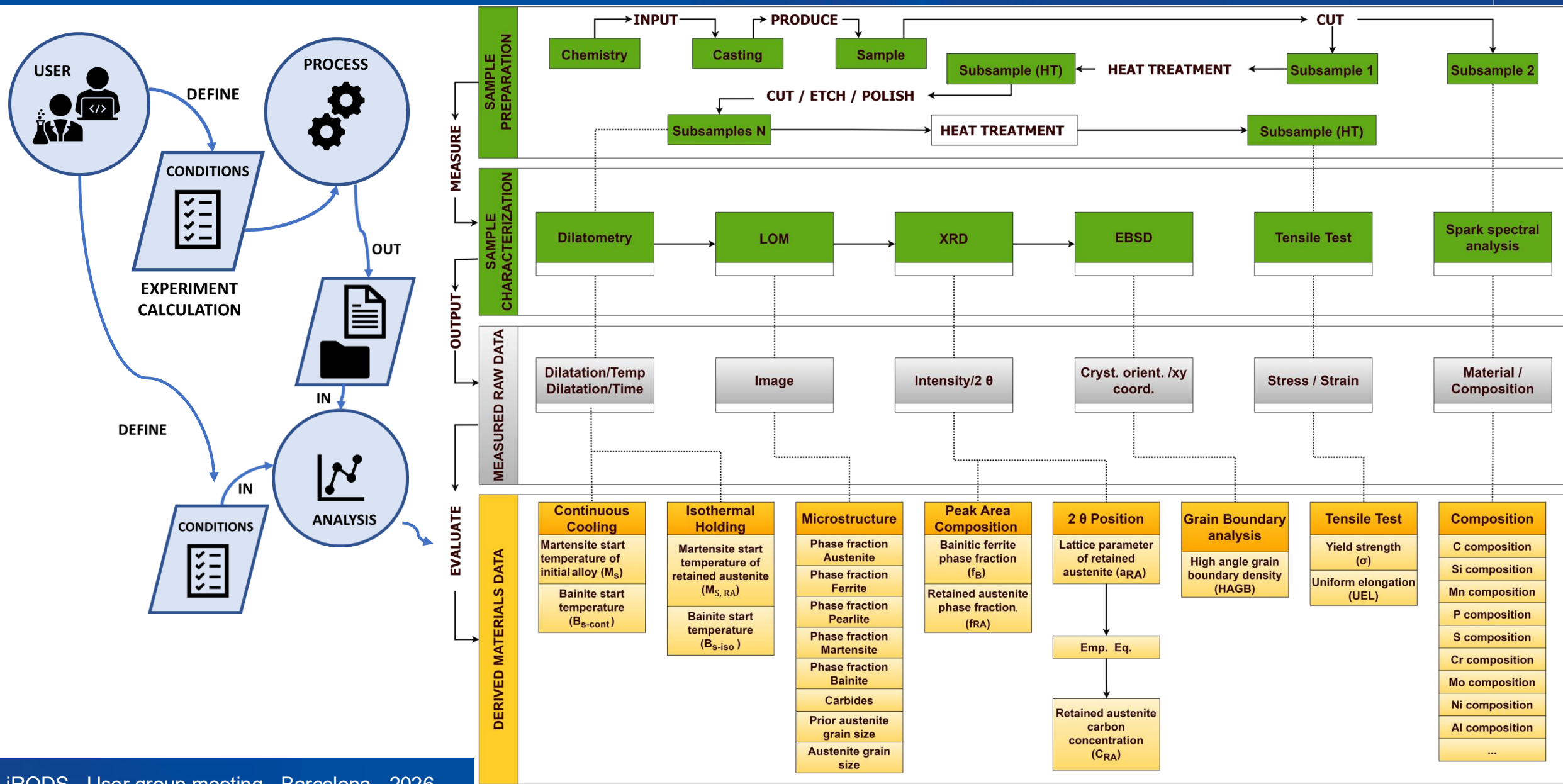


Bainite Use Case





Bainite Use Case





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Alpmat - Materials Center Leoben Forschung GmbH

Active Learning Loop in ALPmat

ALPmat

Favorites

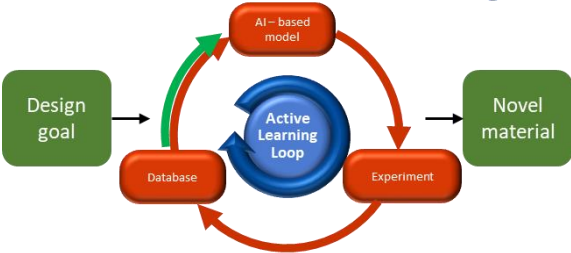
JupyterHub
Perform data analysis using Jupyter notebooks. Python and R are supported; Elyra pipelines available. Data/Metadata can be accessed from iRODS or Literature Search Tools.
#jupyter #python #analysis

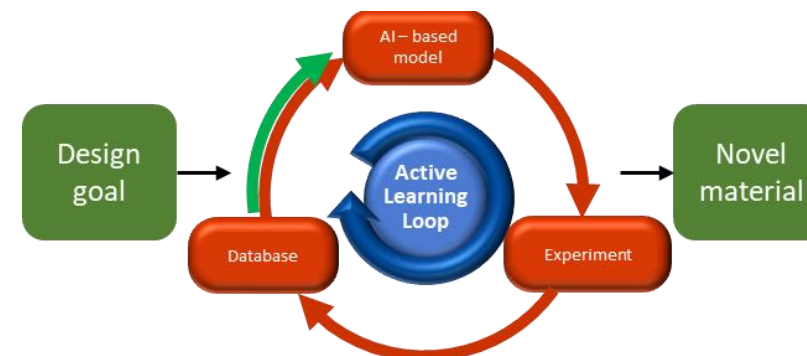
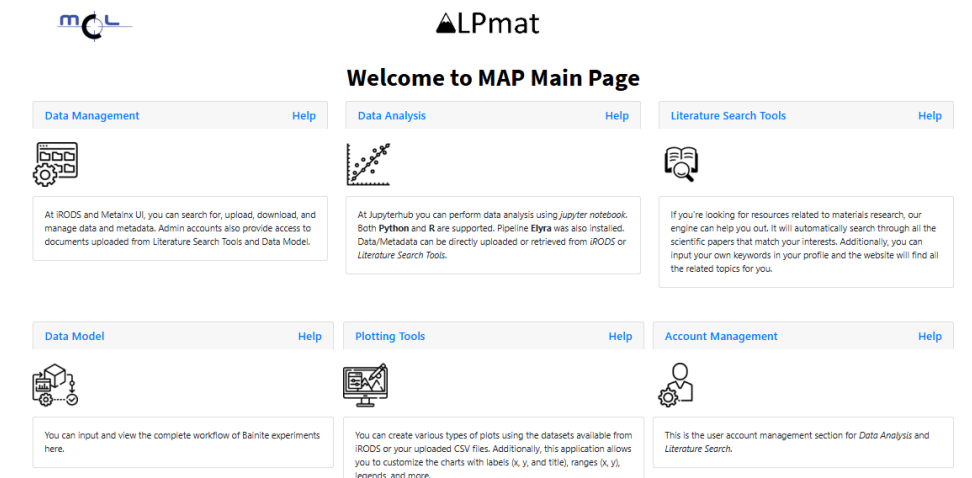
Data Management
At iRODS and Metalnx UI, you can search for, upload, download, and manage data and metadata. Admin accounts also provide access to documents uploaded from Literature Search Tools and Data Model.
#irods #metalnx #data #metadata



USAGE

Framework
Accelerated Materials Design

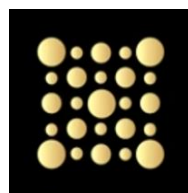




What's next?

Use of ALPmat in upcoming projects

Interoperability: Interface with other tools, platforms, databases



MaterialsCommons4EU - Driving Digital Innovation in Materials Science within Europe

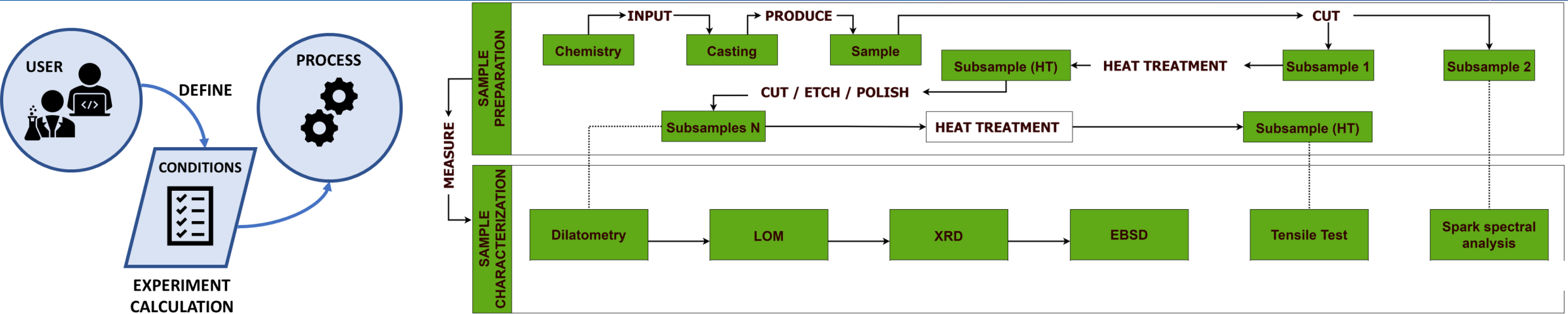
<https://materialscommons4.eu/>



Extend use cases: Data Models for other materials classes, experiments and processes

Integrate ALPmat with Self-Driving Labs

Bainite Use Case



Dilatometry
UID
Subsample UID
Experimentator
TimeStamp
Device
Rod Type
Thermocouple
Thermocouple diam
Extensometer
Cooling medium
Heat Rate

LOM
UID
Subsample UID
Experimentator
TimeStamp
Device

XRD
UID
Subsample UID
Experimentator
TimeStamp
Device
X-ray source
X-ray wavelength
Detector Type
Detector mode
Exposure time per p
Scan Size

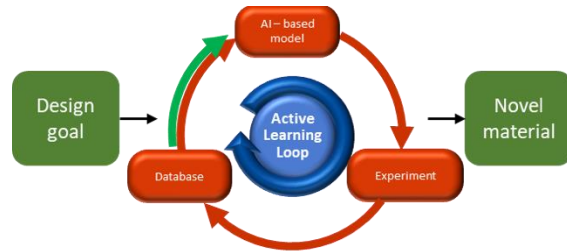
EBSD
UID
Subsample UID
Experimentator
TimeStamp
Device
Detector type
Detector setting
Tilt angle sample
Acceleration voltage
Working distance

Tensile Test
UID
Subsample UID
Experimentator
TimeStamp
Device
Measurement length
Measurement
Test standard
Load speed
Extensometer type
Load cell type

Spark spectral analysis
UID
Subsample UID
Experimentator
TimeStamp
Device
Probe type
Probe
Probe name
Material information
Material standard
Operator



Christian Stecher, Günther Moser, Jürgen Spitaler, N Bedoya (MCL)
Han Tran, Heimo Gursch (Know Center)



Bernd Schuscha, Dominik Brandl, Daniel Scheiber (MCL)
Lorenz Romaner (Montanuniversität Leoben)

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 Bundesministerium
Klimaschutz, Umwelt,
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NEUES DENKEN. NEUES FÖRDERN.


STANDORTAGENTUR
TIROL

Partner of
 **UAR** INNOVATION
NETWORK

